ASSIGNMENT 1 ANSWERS

1. What is data communication? Fundamental components

Data communication is the process of **transmitting data** between two or more devices through a communication medium such as cables or wireless signals.

It involves sending **digital or analog data** from a **sender** to a **receiver** under a set of rules called **protocols**.

Fundamental Components of Data Communication

Below are the **five major components** (with diagrams for notes):

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1. Sender (Source)

The device that **originates** or **sends data**.

Examples:

- Computer
- Smartphone
- Sensor
- Server

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2. Receiver (Destination)

The device that **receives** the transmitted data.

Examples:

- Computer
- Router
- Printer
- Mobile phone

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3. Message (Data)

The actual information that needs to be communicated.

Examples:

- Text
- Audio
- Video
- Documents
- Sensor readings

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4. Transmission Medium

The physical or wireless path through which the data travels.

Types include:

- **Guided media:** Twisted pair cable, coaxial cable, fiber optics
- **Unguided media:** Radio waves, microwaves, infrared

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5. Protocols

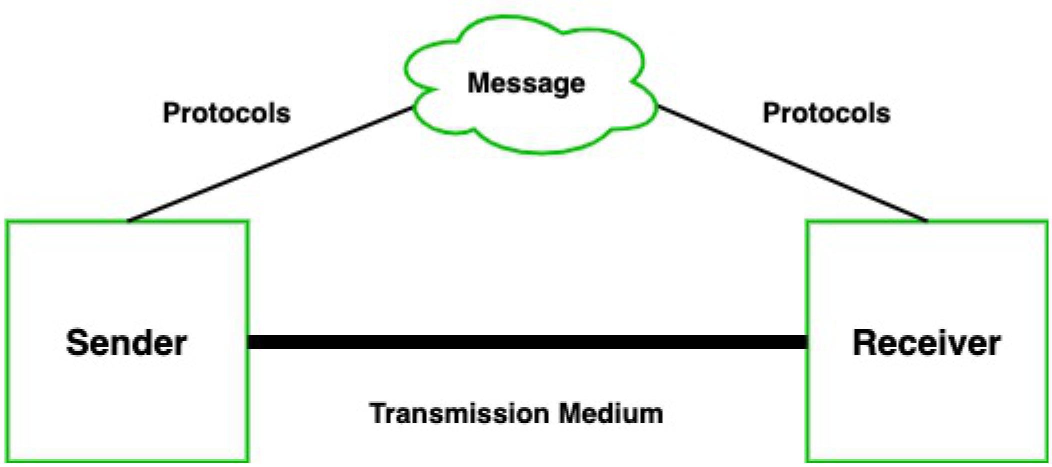
A set of rules that govern:

- How data is formatted
- How it is transmitted
- How errors are detected
- How communication sessions are managed

Examples:

- TCP/IP
- HTTP
- FTP
- SMTP

These components ensure that communication is possible, reliable, and understandable between devices.

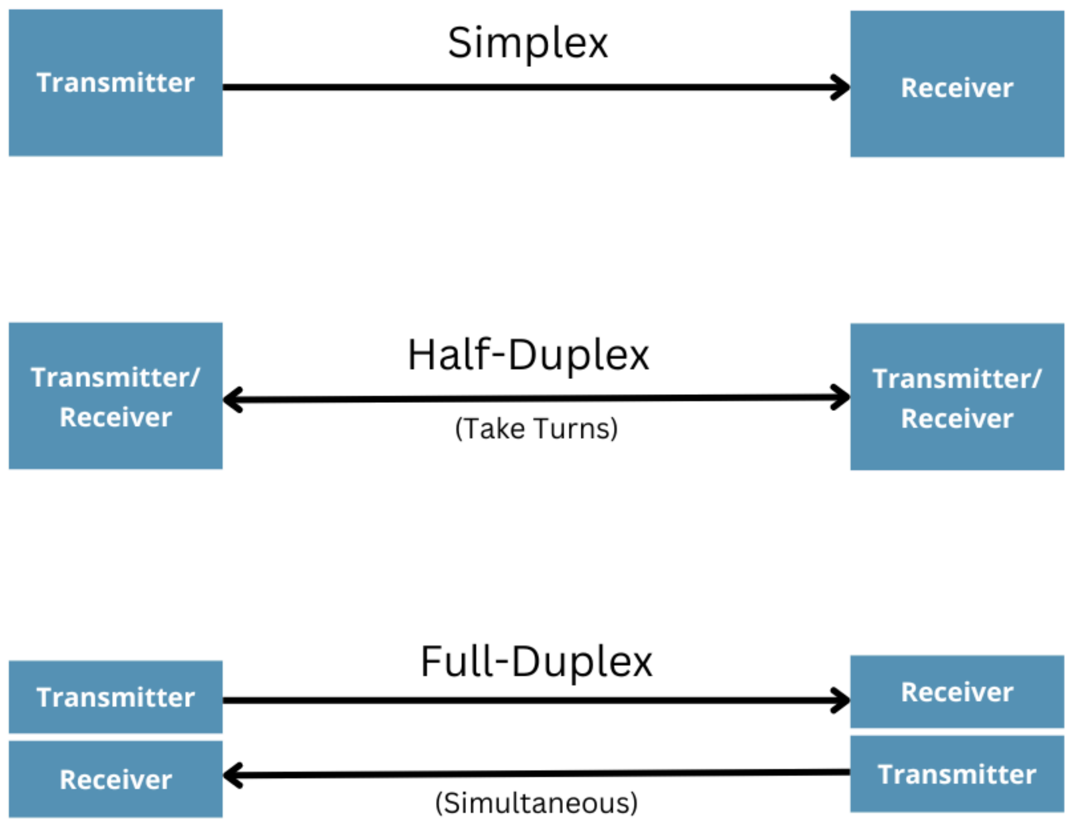


2. Simplex, Half-Duplex, Full-Duplex data flow — with examples

Data flow (or transmission mode) describes the direction(s) in which data can flow between sender and receiver. There are three basic types:

- **Simplex:** Data flows in only one direction — from sender to receiver. Receiver cannot send back to sender.
  - *Example:* A keyboard → CPU connection; a keyboard only sends keystrokes to the computer, not vice versa. Or like a TV broadcast — the station transmits, you only receive.
- **Half-Duplex:** Data can flow in both directions, but not simultaneously. At a time, either one end sends and the other receives, then they can swap roles.
  - *Example:* Walkie-talkies: one person talks while the other only listens; then they switch. OR two-way radio communication in which only one side transmits at a time.
- **Full-Duplex (or Duplex):** Data can flow simultaneously in both directions — both ends can send and receive at the same time.
  - *Example:* A telephone call: both participants can speak and listen simultaneously. Similarly, full-duplex Ethernet or modern network links.

These modes define how communication happens and are chosen based on application need and medium.



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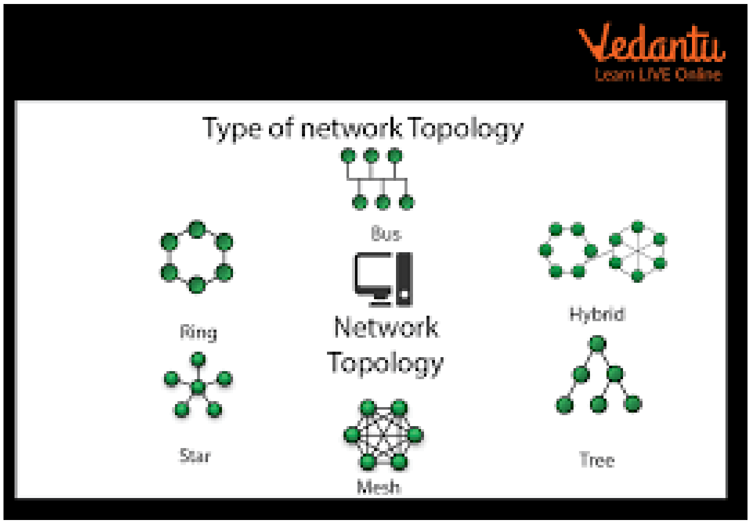
3. Different types of network topologies



Network Topologies – Types, Advantages & Disadvantages

Network topology refers to the **physical or logical layout** of computers, cables, and other network components within a network.

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1. Bus Topology

Description:

All devices are connected to a single central cable called the *bus* or *backbone*.

Advantages:

- Easy and inexpensive to install
- Requires less cable than other topologies
- Good for small networks

Disadvantages:

- Entire network shuts down if main cable fails
- Difficult to troubleshoot
- Performance decreases as more devices are added
- Limited cable length and number of nodes

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2. Star Topology

Description:

All devices are connected to a central device such as a switch or hub.

Advantages:

- Easy to add or remove nodes
- Failure of one device doesn’t affect others
- Easy to troubleshoot
- High performance due to dedicated connection to hub/switch

Disadvantages:

- If the central hub/switch fails, the entire network stops
- Requires more cable than bus topology
- Installation cost is higher

---

3. Ring Topology

Description:

Each device is connected to two neighboring devices, forming a closed loop.

Advantages:

- Data flows in one direction, reducing data collision
- Suitable for high-traffic networks
- Easy to detect faults

Disadvantages:

- If one node or cable fails, entire network is affected
- Difficult to reconfigure or add new devices
- Slower than star topology

---

4. Mesh Topology

Description:

Every device is connected to every other device (fully connected network).

Advantages:

- Provides high redundancy and reliability
- Failure of one link does not affect network
- Very secure due to many communication paths
- Excellent performance

Disadvantages:

- Very expensive (lots of cabling & ports needed)
- Complex installation and maintenance
- Not practical for large networks

---

5. Tree (Hierarchical) Topology

Description:

A combination of star topologies arranged in a hierarchical form.

Advantages:

- Easy to expand
- Centralized monitoring and control
- Fault in one segment does not affect whole network

Disadvantages:

- If backbone line fails, whole network is affected
- Requires more cable
- Complex than star topology

---

6. Hybrid Topology

Description:

A combination of two or more different topologies (e.g., Star + Bus).

Advantages:

- Highly flexible
- Can be designed based on needed performance
- Faults can be isolated easily

Disadvantages:

- Expensive design
- Complex installation and configuration
- Needs skilled administration

You can also consider **logical vs physical topology** — where physical refers to actual wiring layout, and logical refers to how data flows (which may differ from physical layout).

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4. Difference between OSI and TCP/IP models (with diagrams)

Here's a comparison between the two major reference/network models used to understand network communication:

Feature / Aspect	OSI Reference Model	TCP/IP Model (Internet Protocol Suite)
Number of Layers	7 layers: Application → Presentation → Session → Transport → Network → Data Link → Physical	Typically 4 layers (sometimes 5): Application → Transport → Internet (Network) → Link (Network Interface) [some versions include a separate Physical/Network Interface layer]
Purpose	A theoretical, generic model to standardize communication functions in layers — mainly for learning, design, and troubleshooting.	A practical model used in actual networks (like the Internet), grouping protocols and functions used in real-world networking.
Layer Functions / Division	Layers are more granular — separate layers for presentation (data format), session (dialogs, synchronization), etc.	Layers are broader — e.g. no separate Presentation or Session layers; application layer handles application + presentation + session-level tasks. Internet layer handles IP/routing, link layer handles physical + data link aspects.
Common Usage	Good for teaching, protocol design, conceptual clarity.	Used in real-world protocols, Internet communication, modern network implementations.

Diagrammatically:

- OSI: 7 stacked layers. (From lowest: Physical → Data Link → Network → Transport → Session → Presentation → Application)
- TCP/IP: 4-layer (or 5) stack. (Link/Network Interface → Internet → Transport → Application)

Please go through this videos:

https://www.instagram.com/reel/DNX1RSjzUkd/?igsh=MTZzb3J1YXN3eGcxdw%3D%3D

https://www.instagram.com/reel/DSE6LexESUr/?igsh=MWFld3U1eG41eHdheg==

OSI Model

https://www.youtube.com/watch?v=qBXmbJZQ5rY

TCP/IP Model

https://www.youtube.com/watch?v=wvPe4Zb0tUA

OSI vs TCP/IP Comparison

https://www.youtube.com/watch?v=8XHFAc1ZWgk

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5. OSI Reference Model — Functions of each layer

The OSI Reference Model divides the network communication process into **7 layers**, each with specific responsibilities. Here's what each does:

1. **Physical Layer:** Deals with actual physical connection — transmits raw bits (0s and 1s) over physical medium (cables, radio). Deals with signaling, voltage levels, timing, data rates, connectors.

2. **Data Link Layer:** Responsible for node-to-node data transfer on the same medium; frames packets from Network layer into frames, does error detection/correction (via CRC, framing), flow control, physical addressing (MAC), access control to the medium.

3. **Network Layer:** Handles routing of data — logical addressing (like IP addresses), packet forwarding, path determination across different networks (internetworking), fragmentation, and reassembly.

4. **Transport Layer:** Provides end-to-end communication, error detection & correction, flow control, segmentation/reassembly, reliability (or not), multiplexing/demultiplexing. Example protocols: TCP (reliable, connection-oriented) or UDP (unreliable,

connectionless).

5. **Session Layer:** Manages sessions or dialogues between applications — setting up, managing and terminating connections; synchronizing dialogue, checkpoints, recovery, controlling dialogs.

6. **Presentation Layer:** Translates data formats between application and network — encryption/decryption, compression, data translation, serialization, ensures that data is in a common format for sender/receiver.

7. **Application Layer:** The topmost layer — provides network services directly to end-user applications (web browsers, email clients, file transfer, remote login, etc.). Responsible for high-level protocols and data exchange.

So when you send data (say, a file through a messaging app), the data moves from Application layer downward through each layer — each layer adding its own header/trailer (encapsulation), until Physical layer sends the bits. On the receiving end, encapsulation is reversed (decapsulation) — and data is delivered up to the application. This layering helps modularize and simplify network communication.

The OSI Reference Model (Part 2)

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6. Evolution of computer networks; requirements and applications of networks in modern industries

Evolution (at a high level):

- Early computer systems were standalone, isolated — no networking.
- Then came simple data communication (point-to-point) over cables.
- As demand grew, small Local Area Networks (LANs) connected computers within a building.
- With development of protocols like TCP/IP and networking hardware (routers, switches), networks expanded — WANs, the Internet.
- Wireless networks (Wi-Fi, cellular) added mobility.

- Modern networks now support high-speed data, multimedia, distributed computing, cloud, IoT, real-time applications — integrated globally.

Requirements of networks in modern industries:

- **Reliability & Robustness:** Data must be delivered correctly and consistently, even with failures or heavy load.
- **Scalability:** Networks must scale as number of devices/users grows — businesses, cloud services, global reach.
- **Speed & Bandwidth:** High data rates for streaming, large data transfers, real-time apps.
- **Flexibility & Connectivity:** Support heterogeneous devices, remote access, wireless/wired, IoT devices.
- **Security & Privacy:** Secure data transmission, encryption, authentication to protect sensitive data.
- **Interoperability & Standardization:** Use standard protocols so devices from different manufacturers collaborate.
- **Cost-effectiveness:** Efficient resource use, network management, maintenance.

Applications of networks in modern industries:

- **Enterprise & Business Communication:** Email, video conferencing, VoIP, collaborative tools, internal data sharing.
- **Cloud Computing & Data Centers:** Distributed storage, web hosting, SaaS, virtual machines — all depend on networks.
- **Internet Services & E-commerce:** Online shopping, banking, streaming, social media — powered by global networks.
- **IoT (Internet of Things):** Smart factories, sensors, automation, remote monitoring; networks connect and coordinate devices.
- **Telecommunications & Mobile Networks:** Voice, data services, 4G/5G, mobile internet, remote access.
- **Education & Remote Learning:** Online classes, remote labs, global collaboration.

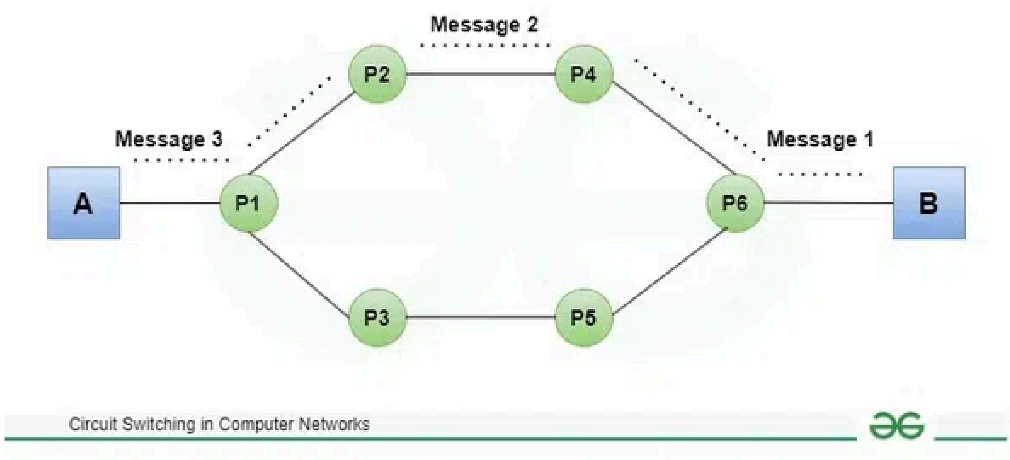
- **Healthcare:** Telemedicine, remote diagnostics, centralized patient records, real-time monitoring.

- **Research & Cloud-based Computation:** Collaboration across countries, data sharing, distributed computing, big data analytics.

Thus — networks became the backbone of nearly every modern industry, enabling communication, collaboration, data exchange, and services at scale.

ASSIGNMENT 2 ANSWERS

1. What are the three phases of circuit switching? (BT-1: Remember)



Circuit switching occurs in three phases:

1. Circuit Establishment

- **A dedicated communication path is created between the sender and receiver.**
- **Resources (bandwidth, channels) are reserved.**
- **Example: Telephone call setup (dialing → ringing → answering).**

2. Data Transfer

- **Actual data is transmitted along the dedicated path.**
- **Constant bandwidth and delay are guaranteed.**
- **No interruptions since the path remains exclusive.**

3. Circuit Termination

- **The connection is released after the communication ends.**
- **All reserved resources are freed for other users.**

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2. Differentiate between datagram and virtual circuit packet switching. (BT-2: Understand)

Feature	Datagram Packet Switching	Virtual Circuit Packet Switching
Path	Each packet takes a different path	All packets follow the same path
Setup Phase	No setup (connectionless)	Requires setup (connection-oriented)
Packet Info	Each packet has full destination address	Only first packet has full address; others use VC number

Reliability	Less reliable, packets may be lost, reordered	More reliable, in-order delivery
Delay	Variable	More stable delay
Example	Internet (IP)	ATM network, Frame Relay

Lec-18: Datagram Switching Vs Virtual Circuit Switching in Pack…

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3. Compare circuit switching and packet switching in terms of efficiency, delay, and reliability. (BT-4: Analyze)

Parameter	Circuit Switching	Packet Switching
Efficiency	Low – dedicated path remains unused during silence	High – bandwidth shared among many users
Delay	Low & predictable (fixed path)	Variable delay (queuing, congestion)
Reliability	Very high once the circuit is established	Depends on packet loss, routing, congestion
Bandwidth Usage	Wasted during no tranmission	Efficient, no reserved bandwidth

Suitability	Real-time, continuous data (voice calls)	Burst data (internet traffic)

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4. Steps of implementing network software using layered protocol architecture (BT-3: Apply)

A layered protocol architecture (like OSI, TCP/IP) organizes communication functions into layers.

Steps:

1. Define the Network Requirements

- **Identify required services (routing, error control, security).**

2. Divide Functions into Layers

- **Each layer should perform a specific set of functions.**
- **Example layers: Application, Transport, Network, Data Link.**

3. Design Protocols for Each Layer

- **Define message formats, rules, timing, error handling.**
- **Ensure protocols are independent of each other.**

4. Implement Layer Interfaces

- **Layers communicate only via well-defined interfaces.**
- **Ensures modularity and ease of updates.**

5. Develop and Test Each Layer Independently

- **Unit testing for protocol behavior.**
- **Verify message encoding/decoding.**

6. Integrate Layers

- **Combine layers and test end-to-end communication.**

7. Deployment & Maintenance

- **Install on network devices.**
- **Monitor performance and update when necessary.**

---

5. With neat diagrams, explain circuit switching and packet switching. Compare both and suggest which is better for data communication. (BT-5: Evaluate)

Circuit Switching (Text Diagram)

```
Sender ----[Path 1]---- [Switch 1] ---- [Switch 2]
---- Receiver
(Dedicated path established before data transfer)
```

Packet Switching (Text Diagram)

```
Packet 1: Sender → Router A → Router C → Receiver
Packet 2: Sender → Router B → Router C → Receiver
(Different paths for each packet)
```

Comparison & Recommendation

Feature	Circuit Switching	Packet Switching
Path	Fixed	Dynamic
Bandwidth	Reserved	Shared
Delay	Predictable	Variable
Efficiency	Low	High
Use Case	Voice calls	Internet data

Which is better for data communication?

✔ **Packet switching is better for data communication because:**

- **Efficient bandwidth usage**

- **Handles bursty traffic**
- **Supports error correction**
- **Scalable for large networks**

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6. Case Study (VoIP Communication): Should the company use circuit switching or packet switching? Justify. (BT-6: Create/Evaluate)

Scenario:

A company wants to set up VoIP (Voice over IP) for office communication.

Evaluation:

Parameter	Circuit Switching	Packet Switching
Cost	High (dedicated lines)	Low
Bandwidth Usage	Wasted during silence	Efficient
Scalability	Limited	Highly scalable
Real-time Support	Good	Good with QoS
Integration with Internet	Poor	Excellent

Recommended Choice: ✔ Packet Switching with QoS

Justification:

- **VoIP runs on IP networks, not on traditional circuits.**
- **Packet switching supports compression, reducing costs.**
- **Quality of Service (QoS) ensures low delay and jitter suitable for voice.**
- **Easy integration with existing LAN/WAN.**
- **Cost-effective and supports remote communication.**

Conclusion:

Packet Switching is the most suitable and efficient method for VoIP implementation.

Assignment 3:

Explain in detail the layered architecture of computer networks. Discuss how encapsulation and de-capsulation occur during data transmission.

Layered Architecture of Computer Networks

Introduction

Computer networks are complex systems involving hardware, software, protocols, and transmission media. To manage this complexity, networking is designed using a layered architecture, where the communication process is divided into multiple layers. Each layer performs a specific function and provides services to the layer above it while receiving services from the layer below it. This approach improves interoperability, scalability, modularity, and ease of troubleshooting.

The two most widely used layered models are:

1. **OSI (Open Systems Interconnection) Model – 7 layers**
2. **TCP/IP Model – 4 layers**

OSI Model – Layered Architecture

1. Physical Layer

- Responsible for the **physical transmission of raw bits (0s and 1s)**.
- Defines hardware specifications such as cables, connectors, voltages, and data rates.
- Examples: Ethernet cables, fiber optics, hubs.

2. Data Link Layer

- Ensures **reliable data transfer** between two directly connected nodes.
- Performs framing, error detection, and MAC addressing.
- Divided into:
    - Logical Link Control (LLC)
    - Media Access Control (MAC)
- Examples: Ethernet, ARP, switches.

3. Network Layer

- Handles **logical addressing and routing** of packets across multiple networks.

- Determines the best path for data delivery.
- Example protocols: IP, ICMP, routing protocols.

4. Transport Layer

- Provides **end-to-end communication** between source and destination.
- Ensures reliability, flow control, and error recovery.
- Example protocols: TCP (reliable), UDP (unreliable but faster).

5. Session Layer

- Manages **sessions or connections** between applications.
- Controls session establishment, maintenance, and termination.
- Examples: NetBIOS session services.

6. Presentation Layer

- Responsible for **data translation, encryption, and compression**.
- Converts data into a format understandable by the receiving system.
- Examples: SSL/TLS, JPEG, ASCII.

7. Application Layer

- Provides **network services to end-user applications**.
- Closest layer to the user.
- Examples: HTTP, FTP, SMTP, DNS.

2. Discuss error detection and correction techniques such as CRC and Hamming Code with examples.

Answer:

- **CRC (Cyclic Redundancy Check):** This is an error detection technique that uses polynomial division to detect errors during data transmission.
- **Hamming Code:** This is an error correction technique that uses additional parity bits to correct single-bit errors in the transmitted data.

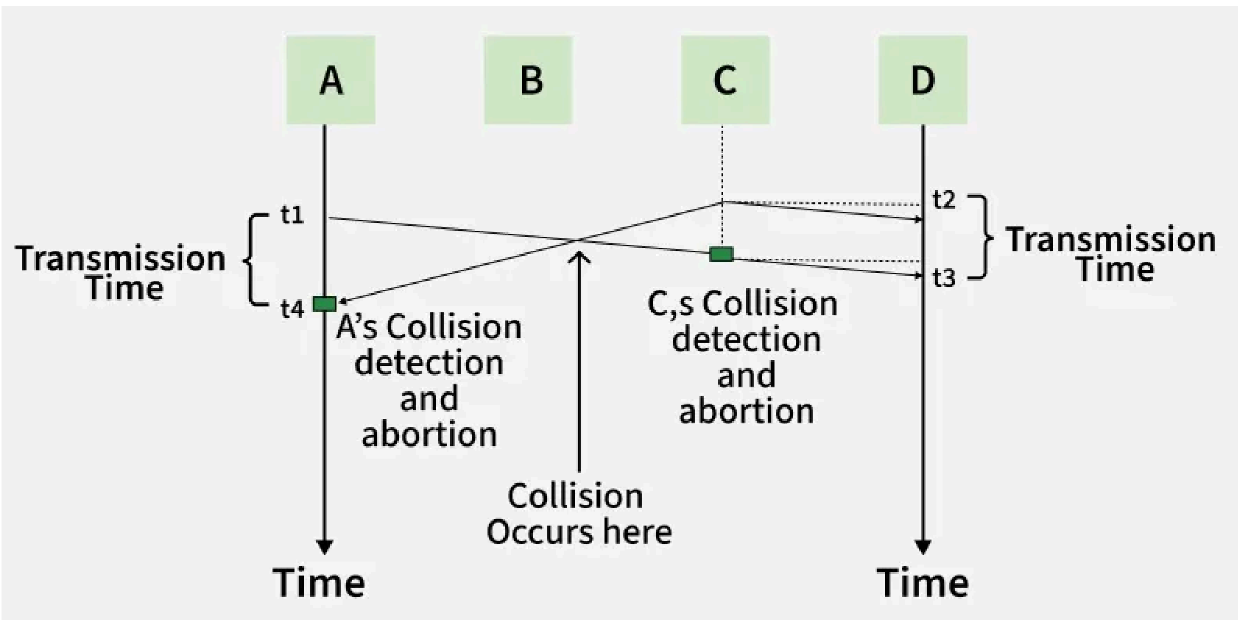
Computer Networks 13 | Hamming Code | CS & IT | GATE Crash…

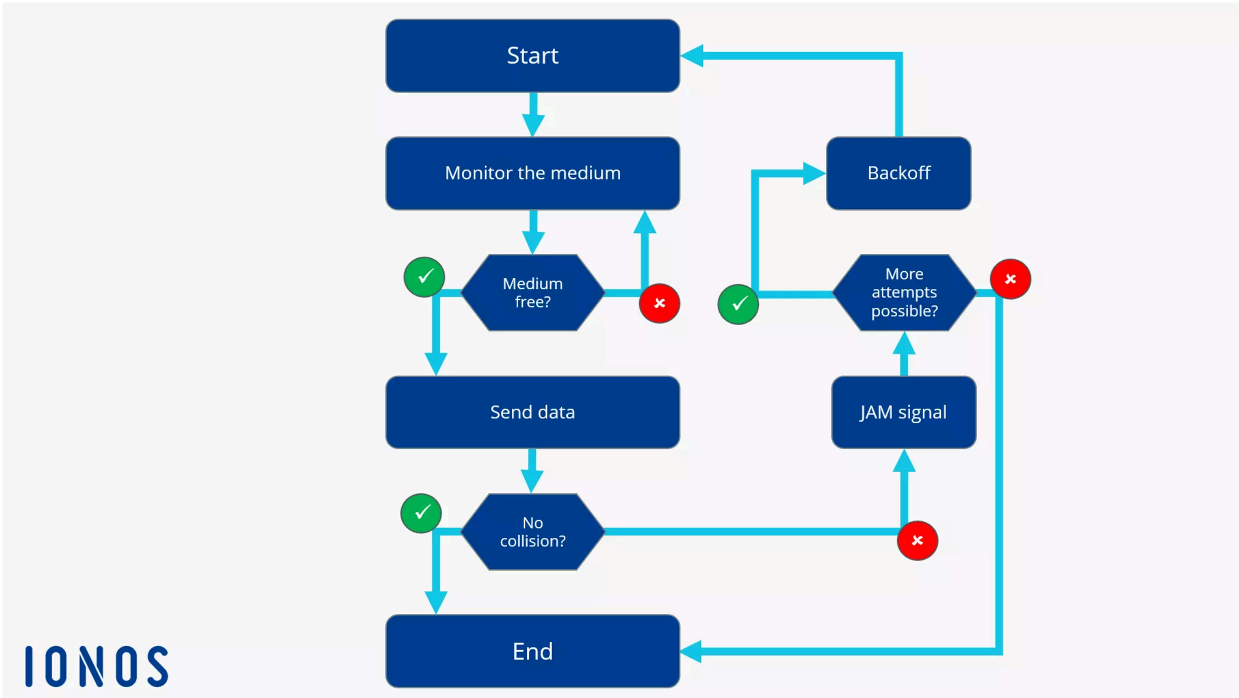
Cyclic Redundancy Check (CRC) - Part 1

3. Explain multiple access techniques such as CSMA/CD, and CSMA/CA & Compare their efficiency.

Answer:

1. CSMA/CD – Carrier Sense Multiple Access with Collision Detection





How It Works

1. A device *listens* to the channel before sending (Carrier Sense).
2. If the channel is idle, it starts transmitting.
3. If two devices transmit at the same time → collision occurs.
4. Devices detect that collision (Collision Detection) and:
    - Immediately stop sending.
    - Send a jam signal to notify others.
    - Wait a random backoff time before trying again.

Key Points

- **Used in wired Ethernet networks (IEEE 802.3).**
- **Works well on shared physical media (e.g., coaxial cable).**
- **Efficient under light to moderate load.**
- **Less effective as the number of devices increases.**

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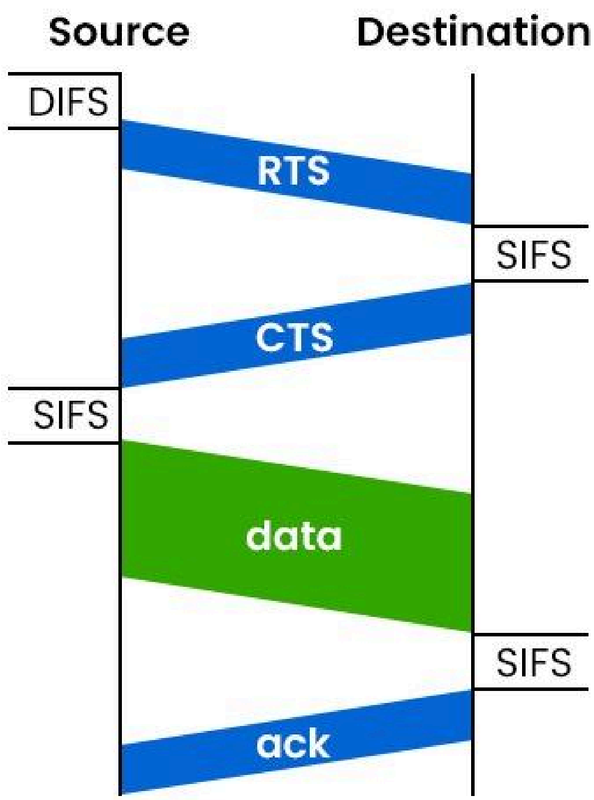
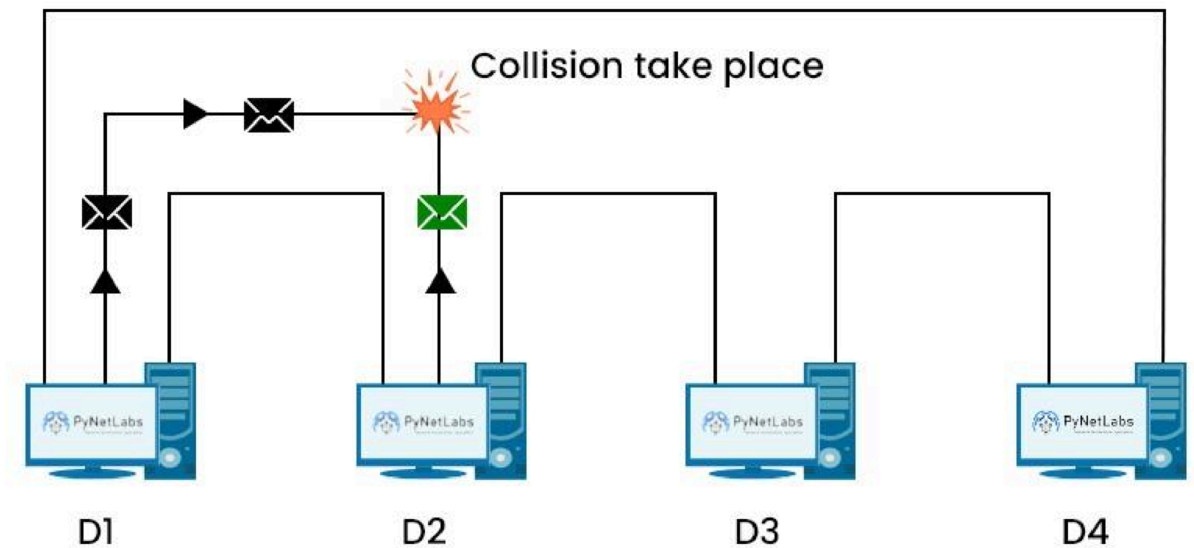
2. CSMA/CA – Carrier Sense Multiple Access with Collision Avoidance



CSMA CA



How It Works

1. **Device listens to the channel (Carrier Sense).**
2. **If the channel is *free*, wait a short time (inter-frame space) then send.**
3. **Uses collision avoidance, not detection:**
    - **Sometimes sends RTS (Request To Send) and waits for CTS (Clear To Send) before sending data.**
    - **After sending data, expects an ACK from the receiver.**

Key Points

- **Used mainly in wireless networks like Wi-Fi (IEEE 802.11).**
- **Wireless devices *cannot listen while transmitting*, so collisions can't be detected reliably — hence avoidance.**
- **Extra control messages (RTS/CTS/ACK) introduce overhead.**

Comparison: CSMA/CD vs CSMA/CA

Feature	CSMA/CD	CSMA/CA
Collision Handling	**Detects collision after it happens**	**Avoids collision before transmission**
Environment	**Wired (Ethernet)**	**Wireless (Wi-Fi)**
Collision Detection	**Yes**	**No**
Overhead	**Lower**	**Higher (RTS/CTS/ACK)**
Efficiency	**Higher under light load**	**Lower due to control overhead**

Suitable Standard	**IEEE 802.3**	**IEEE 802.11**

Efficiency Comparison

- **CSMA/CD is more efficient in wired networks where collisions can be detected and quickly resolved.**
- **CSMA/CA has lower throughput in wireless networks due to collision avoidance overhead (RTS/CTS, ACK waiting).**
- **CSMA/CA is *necessary* in wireless because devices cannot detect collisions while transmitting.**

Please refer this blog for better understanding

Describe the IPv4 addressing scheme and explain the process of sub-netting with an example.

IPv4 Addressing Scheme

Introduction

IPv4 (Internet Protocol version 4) is the most widely used network-layer protocol for identifying devices on a network. It provides logical addressing so that data packets can be routed from the source to the destination across interconnected networks.

An IPv4 address is a 32-bit number, divided into four 8-bit octets, written in dotted decimal notation.

Refer this link , it will give clearity on this topic

6. Explain the implementation and working of Network Address Translation (NAT). Discuss its advantages and limitations

What is NAT? (In Simple Words)

NAT is a technique that allows many devices to use the Internet using only ONE public IP address.

Think of NAT like a receptionist in an office:

- **Inside the office → employees have internal extension numbers**
- **Outside the office → office has one phone number**
- **Receptionist connects calls correctly**

Why Do We Need NAT?

Problem:

- **Internet uses IP addresses.**
- **IPv4 addresses are limited.**
- **Giving one public IP to every device is not possible.**

Solution:

Use private IP addresses inside the network
Use one public IP address to access the Internet

This is done using NAT.

Private IP vs Public IP (Easy)

Private IP (Inside your home/college)

- **Used only inside local network**
- **Examples:**
  - **192.168.1.2**
  - **192.168.1.3**

Public IP (On the Internet)

- **Given by ISP**
- **Visible to the whole Internet**
- **Example:**
  - **103.25.48.10**

Where is NAT Used?

NAT is implemented in:

- **Wi-Fi router**
- **Gateway router**
- **Firewall**

Your home Wi-Fi router already uses NAT.

How NAT Works (Step-by-Step Example)

Suppose:

- **Laptop IP: 192.168.1.2**
- **Mobile IP: 192.168.1.3**
- **Router public IP: 103.25.48.10**

Step 1: Laptop Sends Request

Laptop wants to open Google:

From: 192.168.1.2
To: google.com

Step 2: NAT Changes the Address

Router:

- **Replaces private IP with public IP**
- **Stores information in NAT table**

192.168.1.2 → 103.25.48.10

Step 3: Google Sends Reply

Reply comes to:

103.25.48.10

Step 4: NAT Sends Data Back

Router checks NAT table and sends reply to:

192.168.1.2

Laptop receives correct data

NAT Table (Very Simple View)

Inside Device	Public IP
192.168.1.2	103.25.48.10
192.168.1.3	103.25.48.10

Types of NAT (Easy)

1. Static NAT

- **One private IP → One public IP**
- **Fixed mapping**
- **Used for servers**

Example:

192.168.1.10 ↔ 103.25.48.20

---

2. Dynamic NAT

- **Private IP → Any public IP from a pool**
- **Temporary mapping**

---

3. PAT (Most Common NAT)

Also called NAT Overloading

Many private IPs use ONE public IP
Router uses port numbers to identify devices

✔ This is used in homes, colleges, offices

Advantages of NAT (Easy Points)

Saves public IP addresses
Allows many devices to use one Internet connection
Hides internal devices (basic security)
Cheap and simple to use

Limitations of NAT (Easy Points)

Some online games may not work properly
Video calls and P2P apps can face issues
Hard to access internal devices from outside
Breaks direct device-to-device communication

NAT in Real Life Example

At Home:

- **1 Wi-Fi router**
- **5 mobiles + 2 laptops**
- **Only one Internet connection**

NAT makes this possible

One-Line Summary (For Exams)

NAT allows multiple private IP addresses to share a single public IP address by translating addresses at the router.

ASSIGNMENT 4 ANSWERS

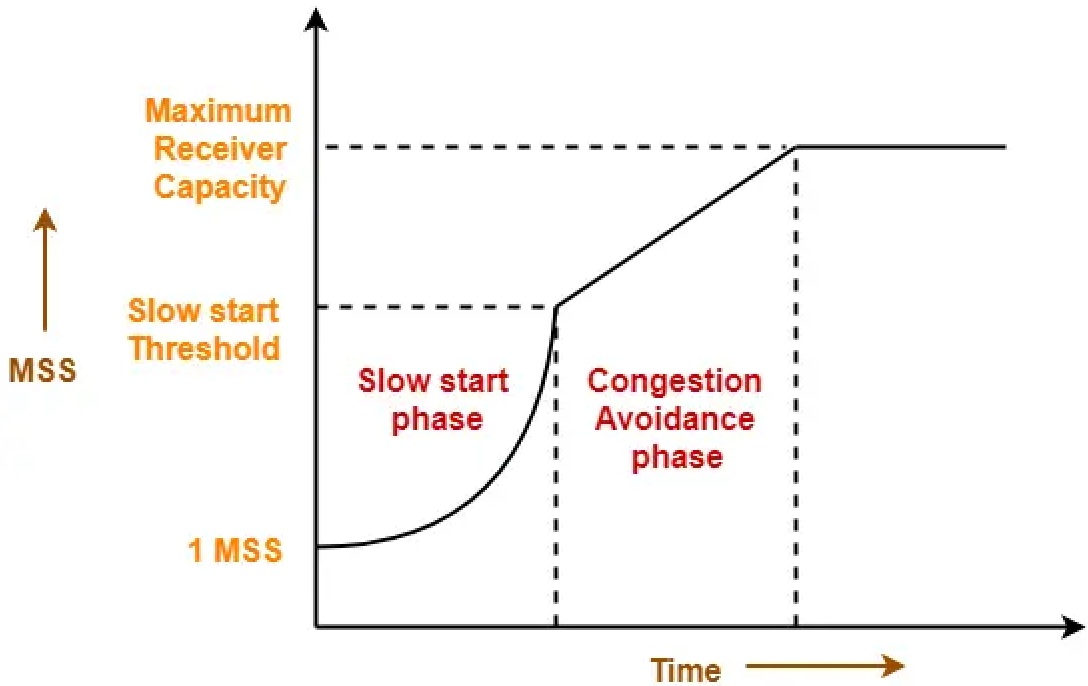
1. Develop a design for congestion control and avoidance in TCP networks. Include queuing mechanisms and QoS

Considerations.Congestion Control in TCP:
Congestion control helps prevent a network from becoming overloaded. It ensures that data packets are sent at a rate the network can handle, thus avoiding data loss, delays, or connection failures. TCP uses several mechanisms for congestion control:

- **Slow Start:** Starts with a small congestion window, gradually increasing the window size as the connection progresses, to avoid overloading the network.

- **Congestion Avoidance:** Uses algorithms like AIMD (Additive Increase, Multiplicative Decrease) to adjust the rate of packet transmission, increasing it gradually, but reducing it sharply if congestion is detected.

- **Fast Retransmit & Fast Recovery:** These mechanisms help TCP recover quickly when a packet is lost due to congestion.



Congestion Control Graph for TCP Protocol

Queuing Mechanisms:
Queuing is used in routers and switches to manage the packets that arrive when the network is congested. It involves organizing packets in different queues based on priority, such as:

- **FIFO (First In, First Out):** Basic queuing where the first packet to arrive is the first to leave.

- **Priority Queuing:** Prioritizes important packets, like voice or video, over regular data.

QoS (Quality of Service) Considerations:
QoS ensures that high-priority traffic, like video conferencing or voice calls, gets the necessary bandwidth and low latency, even during network congestion. It classifies and schedules traffic, making sure critical data is prioritized.

2. Explain in detail the working of application layer protocols HTTP, FTP, and SNMP. Compare their functionalities and use cases.

HTTP (Hypertext Transfer Protocol):

- **Functionality:** HTTP is used for transferring hypertext (web pages) over the internet. It's the foundation of web browsing, where a client (browser) requests resources from a server, and the server responds with the requested data.

- **Use Case:** Browsing websites, requesting web pages, images, videos, etc.

FTP (File Transfer Protocol):

- **Functionality:** FTP is used for transferring files between a client and a server over a network. It allows users to upload and download files from remote servers.

- **Use Case:** Website maintenance, file sharing, and remote data access.

SNMP (Simple Network Management Protocol):

- **Functionality:** SNMP is used for monitoring and managing devices on a network, like routers, switches, and servers. It allows administrators to check device status and make adjustments.

- **Use Case:** Network management, monitoring network devices, and diagnosing issues.

Comparison:

- **HTTP** is for browsing and fetching web pages, **FTP** is for transferring files, and **SNMP** is for managing and monitoring network devices.

- HTTP operates on request-response cycles, FTP on a client-server model, and SNMP uses a manager-agent system.

Learn More:

- HTTP Protocol

- FTP Protocol

- SNMP Protocol

3. Describe in detail the architecture and working of Domain Name System (DNS). Explain recursive and iterative resolution, caching.

What is DNS?

DNS (Domain Name System) is like the **phonebook of the Internet**.

- Humans remember **names** → `google.com`
- Computers understand **numbers (IP addresses)** → `142.250.195.14`

DNS converts website names into IP addresses so your computer can find the correct website.

Imagine if you had to remember phone numbers instead of contact names
Similarly, without DNS, you would have to type IP addresses every time.

Example:

- You type: `www.youtube.com`
- Computer needs: `142.250.xxx.xxx`

DNS does this conversion automatically.

Simple Example

1. You type **www.google.com** in your browser
2. Your computer asks:
   "What is the IP address of google.com?"
3. DNS replies:
   "It is 142.250.195.14"
4. Your browser connects to that IP
5. Google website opens

Advantages of DNS

Easy to remember website names
Fast website access
Works automatically
Scalable for millions of websites

DNS Architecture

Main Parts of DNS Architecture (Easy)

1 User / Client

- You (user) type: `www.google.com`
- Your browser starts the DNS request

Example: Chrome, Firefox, mobile browser

2.Local DNS Server (Resolver)

- First DNS server your computer talks to
- Provided by:
  - Internet Service Provider (Jio, Airtel)
  - Or public DNS (Google 8.8.8.8)

Job:

- Check its memory (cache)
- If IP is known → reply immediately
- If not → ask other DNS servers

Acts like a **middleman**

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3. Root DNS Server

- Top level of DNS hierarchy
- Knows **where `.com`, `.in`, `.org` servers are**

Job:

- Doesn't give IP address
- Says:
  "Ask the `.com` server"

🌍 There are **13 root server systems worldwide**

4.TLD (Top-Level Domain) DNS Server

- Handles extensions like:
  - `.com`
  - `.in`
  - `.org`
  - `.edu`

Job:

- Says:
  "The authoritative server for google.com is here"

---

5. Authoritative DNS Server

- Final and most important server
- Stores **actual IP address**

Job:

- Gives the real IP of the website
- Example:

  google.com → 142.250.195.14

DNS Resolution Process

- When you type a website address, a **DNS resolver** (often your Internet Service Provider's DNS server) takes the request and starts the process of resolving it.
  - **Recursive Resolution:** The resolver takes full responsibility for getting the answer. If it doesn't have the IP address, it will keep asking other DNS servers until it finds the answer.
  - **Iterative Resolution:** In this case, if the resolver doesn't have the answer, it returns the best answer it knows (often the address of a TLD server) and leaves the rest to the client.

DNS Caching

- **Caching** helps speed up DNS resolution by **storing the IP addresses** temporarily. Once the DNS resolver gets the IP, it saves it for a period (TTL or time-to-live) so it doesn't need to ask the DNS servers again.
    - **Caching at different levels:**
        - **DNS Resolver Cache**: Stores answers locally for faster lookup.
        - **Local Browser Cache**: Stores recently visited websites on your device.

5. Evaluate congestion avoidance mechanisms such as leaky bucket and token bucket algorithms. How do they ensure QoS?

What are Congestion Avoidance Mechanisms?

Congestion avoidance mechanisms are used to **control traffic flow** in networks to avoid **overloading** (or congestion), which could lead to delays, packet loss, and poor performance.

Two important mechanisms are:

Leaky Bucket Algorithm

- **How it works**: The leaky bucket algorithm controls the **rate** at which data packets are sent. Think of it like a **bucket** with a small hole at the bottom:
  - **Incoming packets** (data) fill the bucket.
  - The packets leak out **at a fixed rate** from the hole.
  - If the bucket fills up (too much data comes in too fast), it **overflows** and the extra packets are **discarded**.
- **Why it helps?**: It ensures data is sent at a **constant rate**, preventing sudden bursts of traffic that can cause congestion.

Token Bucket Algorithm

- **How it works**: The token bucket algorithm allows bursts of traffic but **limits the average rate**.
  - Tokens are added to the **bucket** at a **constant rate**.
  - To send a packet, a **token** must be available in the bucket. If no token is available, the packet must wait.
  - If the bucket is full, the extra tokens are **discarded**, but the data can still be sent if a token is available.
- **Why it helps?**: This allows **bursty traffic** (short-term increases in data flow) but still controls the **long-term rate** of packet transmission.

How Do They Ensure QoS?

- **QoS (Quality of Service)** ensures that important data (like video calls or gaming) gets through the network with **less delay and jitter** (variance in packet arrival times).
- Both algorithms **control traffic** by smoothing out data flow, which reduces congestion and ensures that **high-priority traffic** is not delayed.

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6. Explain the architecture of the Domain Name System (DNS). Discuss its role in resolving names to IP addresses.

DNS Architecture

The **Domain Name System (DNS)** has a **hierarchical** structure, which means it works in **levels**:

1. **Root DNS Servers**: These are at the top of the hierarchy. They don't know all the IP addresses but know where to direct requests for **Top-Level Domains (TLDs)** (like .com, .org, .net).

2. **TLD DNS Servers**: These servers manage domain names like **.com**, **.org**, etc. They tell you where to find the **Authoritative DNS Servers** for a specific domain.

3. **Authoritative DNS Servers**: These servers know the **exact IP address** for a domain (like www.example.com). They provide the **final answer** to your request.

How Does DNS Resolve Names to IP Addresses?

- When you type a website name (e.g., **www.example.com**), your computer asks the DNS server for the **IP address**.

- The DNS resolver checks its **cache** (a saved list of IP addresses) first. If it doesn't have it, it **asks other servers** (Root → TLD → Authoritative) until it finds the IP.

- Once the IP is found, it's returned to your browser, and the website loads.

DNS Caching:

- DNS **caches** the IP addresses to make future lookups faster. Once a domain's IP is found, it can be stored temporarily (based on **TTL**, or Time To Live) so you don't have to ask again.

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7. Analyze congestion control mechanisms used in TCP. How does TCP maintain reliability and flow control during heavy traffic?

TCP Congestion Control

- **TCP** uses special **algorithms** to prevent congestion and keep the network running smoothly, even during **heavy traffic**.

How Does TCP Maintain Reliability and Flow Control?

1. **Slow Start**:

    - When a TCP connection starts, it **slowly increases** the speed at which it sends data to avoid congestion. It begins with small packets and grows bigger if the network is doing well.

2. **Congestion Avoidance (AIMD)**:

    ○ **AIMD (Additive Increase, Multiplicative Decrease)** is a method used to increase the data flow slowly. When congestion is detected, it **reduces** the sending speed.

3. **Fast Retransmit & Fast Recovery**:

    ○ If a packet is lost, TCP will **immediately retransmit** the missing packet without waiting for a timeout. This **reduces delays**.

4. **Flow Control (Windowing)**:

    ○ TCP uses **windowing** to control the **amount of data** sent before receiving an acknowledgment. This ensures the sender doesn’t overwhelm the receiver.

    ○ The **window size** changes based on the receiver’s capacity and network conditions.

8. Design an IP addressing scheme for a university with four campuses interconnected by routers. Allocate subnets for each department, explain classless addressing (CIDR), and illustrate routing between subnets using a sample routing table.

What is IP Addressing and Subnetting?

IP addressing is the system that assigns unique addresses to devices on a network. **Subnetting** divides a larger network into smaller **subnets** to make network management easier and more efficient.

In this case, we need to design an **IP addressing scheme** for a university with **four campuses**, each with different departments. We will also use **Classless Inter-Domain Routing (CIDR)**, a method of allocating IP addresses without using traditional class-based rules (Class A, B, C).

Steps for Designing the IP Addressing Scheme:

1. **Choose a Private IP Address Range:**
   We will use a private IP address range (since it's for internal use) such as **10.0.0.0/8** (which gives us a large address space).

2. **Subnetting for Four Campuses:**

   - Divide the address space into **subnets** for each campus and department.

   - Use CIDR to allocate subnets. For example, **10.0.0.0/16** can be split into smaller **/24** subnets for departments.

3. Example Subnet Breakdown:
    - **Campus 1:**
        - Department A: 10.0.0.0/24
        - Department B: 10.0.1.0/24
    - **Campus 2:**
        - Department A: 10.0.2.0/24
        - Department B: 10.0.3.0/24
    - Repeat this process for **Campus 3** and **Campus 4**.
4. **Routing Between Subnets:**
    - Use **routers** to route traffic between the subnets across campuses.
    - A **routing table** will define which subnets can reach which other subnets, and how traffic is forwarded.

Sample Routing Table:

Destination Network	Subnet Mask	Next Hop IP Address
10.0.0.0/24	255.255.255.0	192.168.1.1
10.0.1.0/24	255.255.255.0	192.168.1.2
10.0.2.0/24	255.255.255.0	192.168.2.1
10.0.3.0/24	255.255.255.0	192.168.2.2

CIDR Explanation:

- **CIDR (Classless Inter-Domain Routing)** allows for more flexible allocation of IP addresses.
- Instead of the traditional **Class A/B/C**, CIDR uses **subnet masks** (e.g., /24) to define the size of a subnet.
- **Example**: **10.0.0.0/24** means **10.0.0.0** is the network address, and the **/24** means there are 24 bits for the network part (leaving 8 bits for hosts in the subnet).

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9. Explain with a diagram the working of HTTP and SMTP protocols. Compare their roles in data communication.

What is HTTP?

- **HTTP (HyperText Transfer Protocol)** is used to request and transfer **webpages** from a **web server** to a **web browser**.

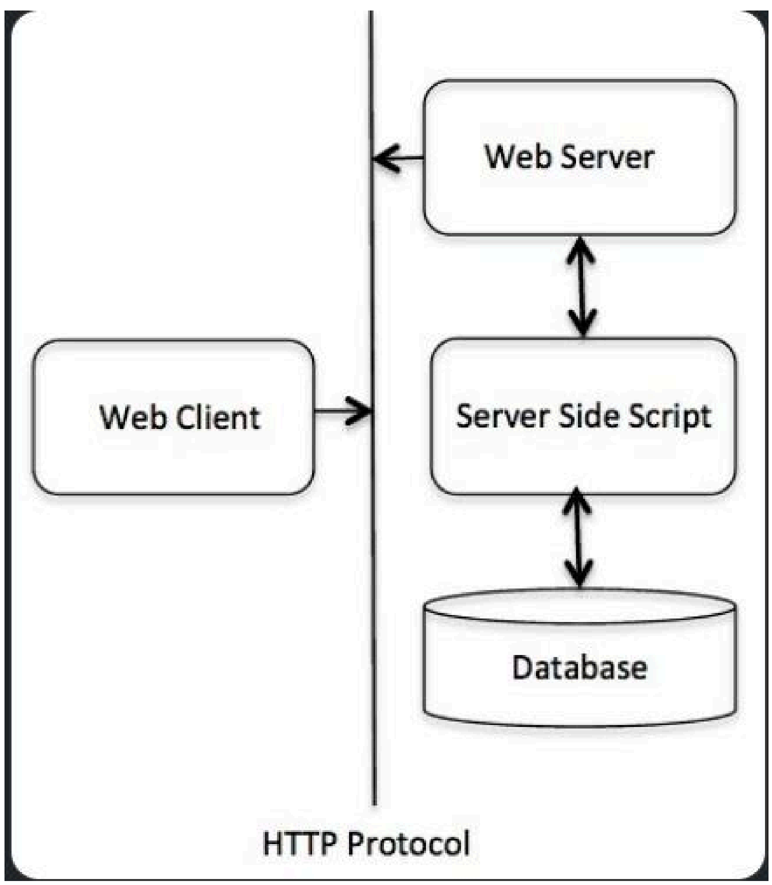
What is SMTP?

- **SMTP (Simple Mail Transfer Protocol)** is used to send **emails** between mail servers. It handles the sending and routing of messages.

How Do HTTP and SMTP Work?

HTTP (Web Browsing)

1. **User enters a website**: e.g., www.example.com
2. **HTTP Request**: The browser sends a request to the web server.
3. **Web server Response**: The server sends back the requested webpage (HTML, images, etc.).
4. **Display on Browser**: The browser displays the webpage to the user.



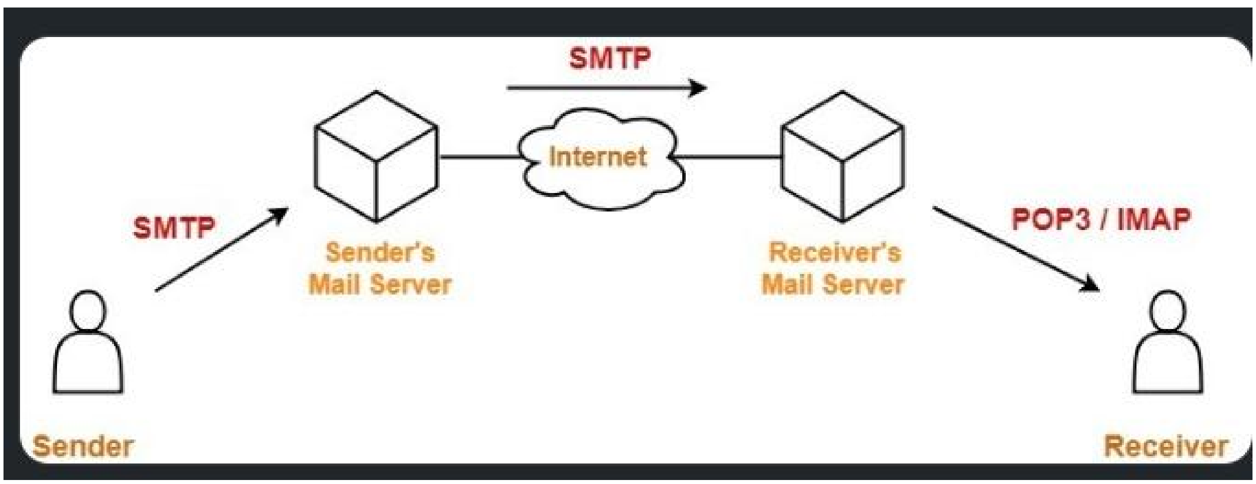
SMTP (Email Communication)

1. **User sends an email** from an email client (e.g., Gmail).

2. **SMTP Request**: The email client connects to the **SMTP server** and sends the email.

3. **Email Delivery**: The SMTP server forwards the email to the recipient's email server.

4. **Recipient Receives Email**: The email is delivered and can be read by the recipient.



HTTP is used for transferring **webpages** from a server to a client.

- **SMTP** is used for sending **emails** from a client to a server.

Feature	HTTP (Hypertext Transfer Protocol)	SMTP (Simple Mail Transfer Protocol)
Primary Role	Used to access websites and transfer web content (text, images, multimedia) [1] [10].	Used for sending and delivering electronic mail between servers [5].
Transmission Model	**Pull Protocol:** The client requests (pulls) information from the server. The server responds only when asked [2].	**Push Protocol:** The client pushes the message to the server, and the sending server pushes it to the receiving server [8].
State Management	**Stateless:** The server does not retain information about the client between requests [2].	**Stateful (Session-based):** Maintains a conversation using specific command sequences (HELLO, DATA, QUIT) until the connection is terminated [9].
Port Number	Uses Port **80** [10].	Uses Port **25** [11].
Data Handling	Transfers content to a device for immediate viewing or processing (e.g., web pages) [5].	Transfers messages to a mail server for storage until the recipient retrieves them [12].